

LORENZO GABRIEL STIGLIANO

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EDUCATION

University of Oxford

D.Phil. Applied Machine Learning

Oct 2025 - Oct 2029

Oxford, UK

- Funded by the Engineering and Physical Sciences Research Council (EPSRC) Center for Doctoral Training in Healthcare Data Science.
- **Modules:** Computational Statistics, Machine Learning, Data Engineering and Infectious Disease Analytics.

Imperial College London

M.Sc. Computing (Artificial Intelligence & Machine Learning), Distinction (79%)

Oct 2022 - Oct 2023

London, UK

- **Dissertation:** *Reproducible Knowledge Distillation for Graph Neural Networks*
- **Modules:** Reinforcement Learning, Maths for Machine Learning, Computer Vision, Deep Learning, Natural Language Processing, Robot Learning, Machine Learning for Imaging.

University of Edinburgh

B.Sc. Computer Science and Mathematics, First Class Honours (84%)

Sep 2017 - Jun 2021

Edinburgh, UK

- **Dissertation:** *Introduction to Inverse Problems and Markov Chain Monte Carlo Methods*
- **Modules:** Machine Learning, Statistics, Database Systems, Algorithms and Data Structures.

WORK EXPERIENCE

Data Scientist

J.P. Morgan Chase & Co.

Sep 2023 - Sep 2025

London, UK

- Developed from scratch an end-to-end machine learning pipeline for rare event prediction, employing a comprehensive approach encompassing classification models and variational autoencoders.
- Optimized RAG pipelines using DSPy and LangChain, enabling cost-effective analysis of over 1000 transcripts per week with a 60% reduction in resource expenditure.
- Implemented data processing pipelines and back-testing frameworks for scalable prototyping using PySpark.
- Leveraged knowledge in Python (NumPy, Pandas, Scikit-Learn, PyTorch, SHAP), SQL, Git, Bash, Spark, MLFlow.

Data Scientist Intern

J.P. Morgan Chase & Co.

Jun 2022 - Aug 2022

London, UK

- Spearheaded data analytics initiative exploring web data to extract actionable insight for 1M+ companies.
- Optimized ETL pipeline with improved throughput, scalability, and cost efficiency.
- Leveraged knowledge in Python (NumPy, Pandas, Matplotlib, SciPy, SQLAlchemy), SQL, Git, Bash, Spark.

Software Engineer

Barclays

Aug 2021 - Jun 2022

Glasgow, UK

- Developed a full-stack web application to evaluate the impact of release changes on the current system.
- Implemented and optimized queuing systems reducing background processing run-times by 50%.
- Leveraged knowledge in Java (Spring), Javascript (React), Rest APIs, HTML/CSS, SQL, Git.

Software Engineering Intern

Barclays

Jul 2020 - Aug 2020

Glasgow, UK

- Created an application to display client relations within the bank using AWS to reduce hosting costs by 90%.
- Leveraged knowledge in Java (Spring), AWS (EC2, RDS, Auto-Scaling), Salesforce API, Rest APIs.

Software Engineering Intern

Imagination Technologies

Jun 2019 - Jul 2019

Kings Langley, UK

- Developed a full-stack app which eliminated the need for manual interactions with hardware emulator platforms.
- Leveraged knowledge in Python (Flask, SQLAlchemy), JavaScript, Rest APIs, HTML/CSS, Git, Confluence.

ACADEMIC EXPERIENCE

Graduate Research Assistant

Apr 2023 - Sep 2023

Brain And Signal Research & Analysis lab, Imperial College London

London, UK

- Developed a novel one-to-many knowledge distillation algorithm for graph neural networks , achieving $\geq 95\%$ reduction in model parameters while maintaining performance in graph classification tasks.
- Algorithm reduced inference times by $\sim 90\%$ on distilled models deployed on edge devices.
- Under the supervision of Dr. Islem Rekik.
- Leveraged knowledge in Python (PyTorch, PyTorch Geometric, NumPy), Git, Bash, and SLURM.

Undergraduate Research Assistant

Jun 2020

Center for Speckled Computing, University of Edinburgh

Edinburgh, UK

- Researched potential causal links between biomarkers and PM2.5 exposure across diverse test groups, employing statistical tests.
- Identified significant differences in subjects exposed to second-hand smoke versus non-exposed individuals.
- Under the supervision of Prof. Damal Arvind.
- Leveraged knowledge in Python (Scikit-learn, NumPy, SciPy), Git and Bash.

TEACHING

- Foundations of Data Science (INFR08030), University of Edinburgh *Spring 2021*
- Introduction to Computation (INFR08025), University of Edinburgh *Fall 2020*
- Algorithms, Data Structures, Learning (INFR08009), University of Edinburgh *Spring 2020*

AWARDS AND PRIZES

- Fully Funded Doctoral Studentship EPSRC Center for Doctoral Training. *Oct 2025*
- University of Edinburgh Class Prize for Top Performance in Computer Science and Mathematics. *May 2021*
- King's College Class Prize for Mathematics, awarded for exceptional performance in mathematics. *June 2017*
- King's College Honorary Scholarship, awarded for outstanding performance at AS Level. *June 2016*

SKILLS

Languages: (*Proficient*) Python, Java, SQL, Bash (*Familiar*) R, Matlab.

Technologies/Frameworks: NumPy, Pandas, Scikit-Learn, PyTorch, PyTorch Geometric, PySpark, MLFlow.

Software Development: Git, Bitbucket, Databricks, Agile, Unit Testing, RESTful APIs, AWS.

Spoken Languages: English (fluent), Spanish (native), Italian (conversational).
